

Course Enhancement Project

Visualizing The Central Limit Theorem Through repeated sampling of Housing Prices

Prepared By:

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Section: G

1-Introduction:

1.1- Objective of CEP:

The main objective of this project is to understand and visualize the Central Limit Theorem (CLT) using real-world housing price data. Instead of only studying the concept theoretically, this project focuses on applying it through a computational approach. By repeatedly drawing random samples from the dataset and calculating their means, we aim to observe how the behaviour of these sample means changes as the sample size increases.

This process helps in building an intuitive understanding of how randomness works in data and how sample means become more stable over time. The project also emphasizes manual calculation of statistical measures such as mean, variance, and standard deviation to strengthen our understanding of the underlying formulas. Overall, the goal is to connect statistical theory with practical implementation and visualization.

1.2- Importance of CLT:

The Central Limit Theorem is important because it helps us deal with real-world data, which is usually messy and not normally distributed. It shows that when we take **many samples and calculate their averages, the results start to form a normal** distribution as the sample size increases.

This makes analysis easier and more reliable, which is why averages are widely used in surveys, experiments, and research. It also helps us understand that larger samples give more accurate results. Overall, CLT is essential because it allows us to make meaningful conclusions even when the original data is not perfect.

2- Dataset Description:

The dataset used in this project is **Indian House Prices**, and only the **Price column was selected for analysis**. This column represents the prices of different houses, making it suitable for studying real-world numerical data.

Before starting the analysis, the data was cleaned by removing missing, zero, and invalid values. Any non-numeric entries, such as symbols or commas, were also handled to ensure that only valid numerical values remained. This step was necessary to make the data accurate and reliable for further calculations.

3-Formulas Used:

- **Mean:** $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$
- **Variance:** $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$
- **Standard Deviation:** $s = \sqrt{s^2}$
- **Standardized Value (Z-score):** $z = \frac{\bar{x} - \mu}{\sigma / \sqrt{N}}$

3-Original Data Distribution:

3.1 Observation:

1. Is the original price distribution symmetric or skew symmetric?
The distribution is extremely right-skewed (positive skew). The value 11.17 is far above 0, indicating very strong asymmetry.
2. Does the original price distribution look normally distributed?

Absolutely not. A normal distribution has a skewness of 0. Your value of 11.17 is massive, meaning the data is far from normal.

3. Why might real-world housing price data be skewed?

i- No upper limit:

Cheap houses have a floor (can't be free), but expensive houses have no ceiling.

ii- Few rich buyers, many middle-class buyers:

Most people buy normal-priced homes. Only a few buy luxury mansions.

iii- Outliers pull the average:

A 10 crore house is rare, but it stretches the tail far to the right.

iv- Location matters:

One beachfront villa can cost 100x more than a similar-sized house 10 km away.

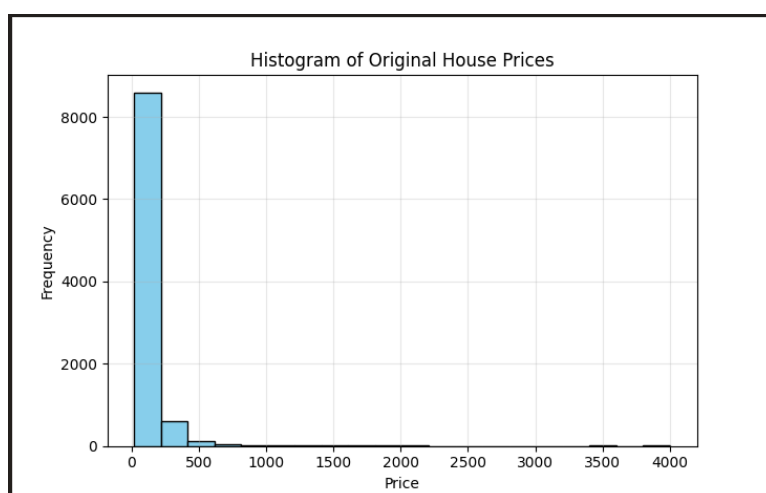
v- Data excludes slums

Very cheap houses are often not recorded, so the left side gets cut off.

3.2 **Data analysis:**

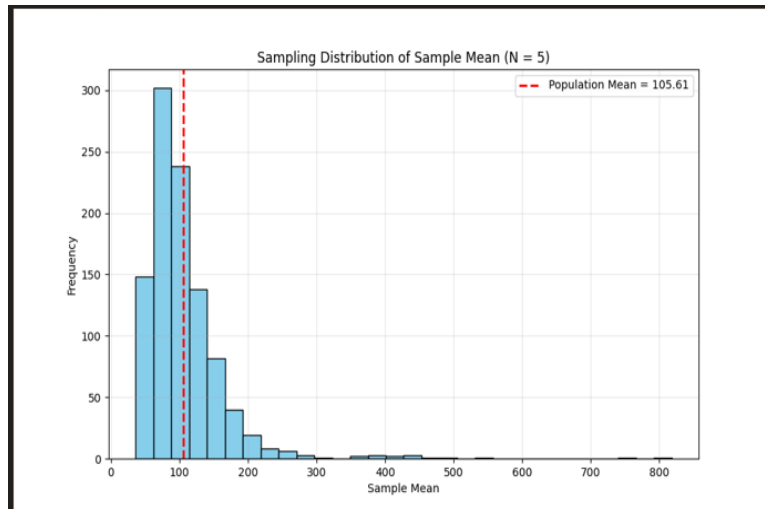
1. Number of valid observations: 9390
2. Minimum price: 20.0
3. Maximum price: 4000.0
4. Manually calculated mean: 105.61
5. Manually calculated variance: 19506.05
6. Manually calculated standard deviation: 139.66

3.3 **Original Data Histogram:**

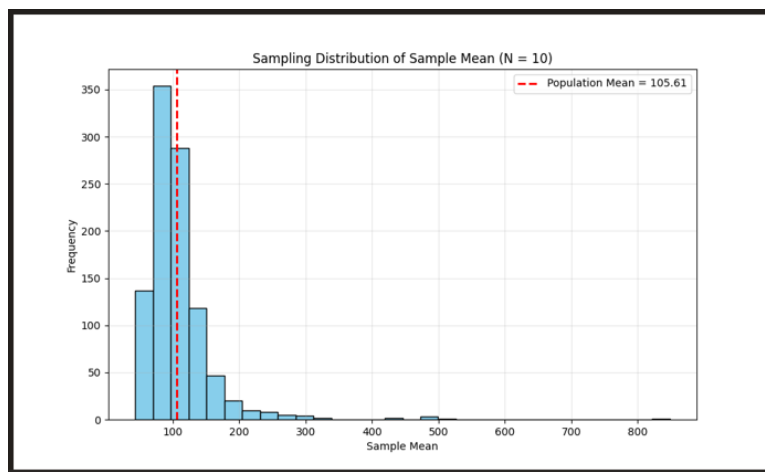


4- Histograms of Sample Means:

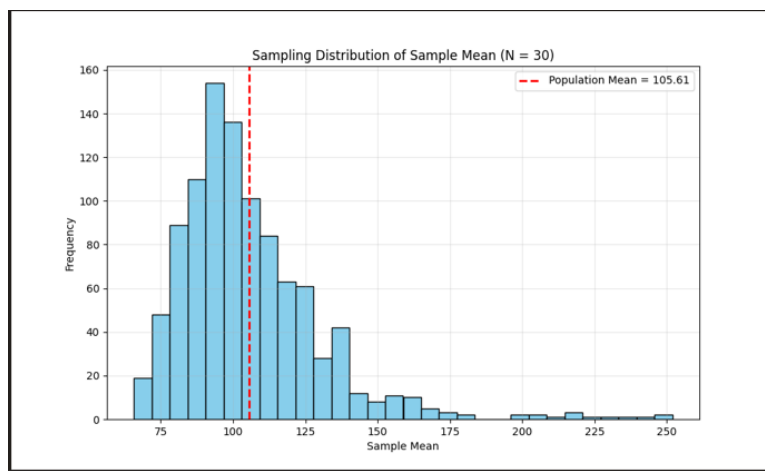
4.1 N=5:



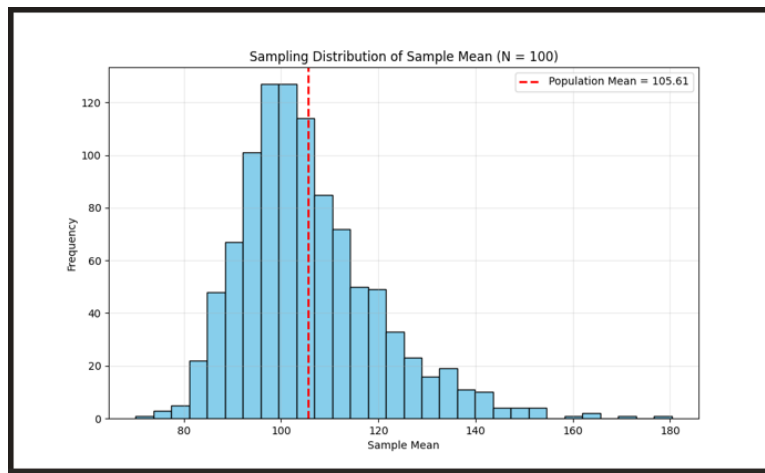
4.2 – N=10:



4.3 – N=30:

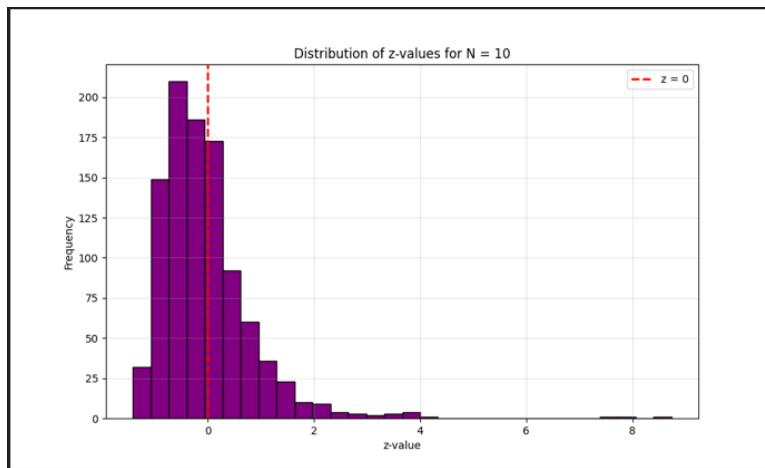


4.4 – N=100:

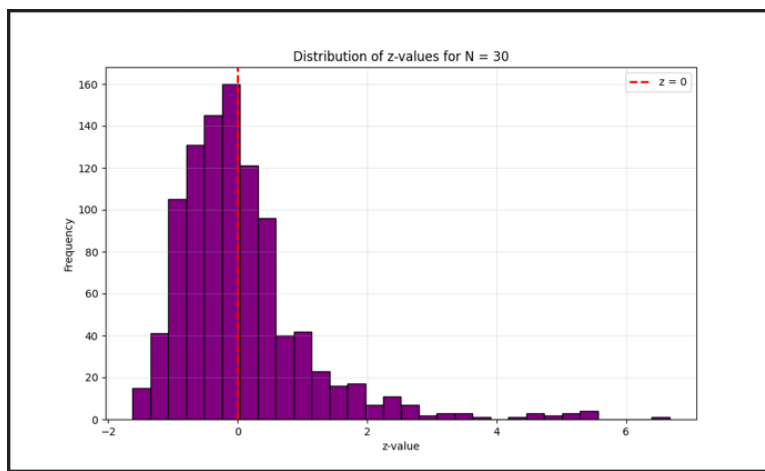


5-Histograms of Standardized Z-Values:

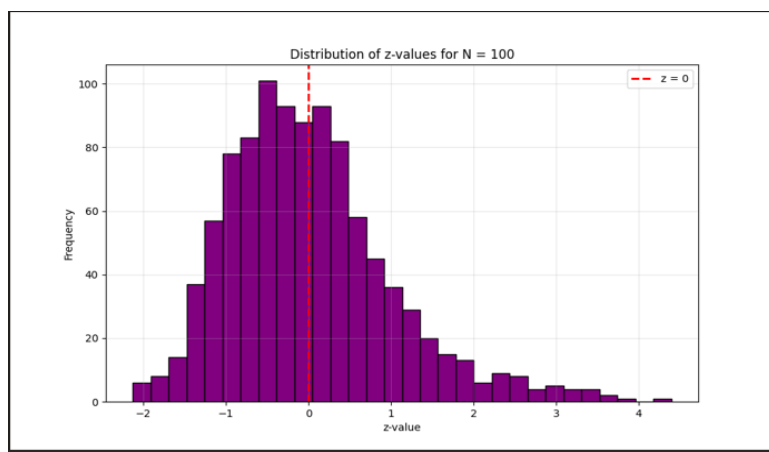
5.1- N=10:



5.2- N=30



5.3- N=100:



6-Observations:

6.1. What happens to the spread of the sample means as N increases?

The spread decreases as N increases. With N=5 the histogram of sample means is very wide, while at N=100 it is much narrower. This is explained by the standard error formula: $SE = \sigma / \sqrt{N}$. A larger N gives a smaller SE, so sample means cluster more tightly around the population mean.

6.2. What happens to the shape of the sampling distribution as N grows?

The shape becomes increasingly symmetric and bell-shaped (normal). At N=5 it is still noticeably skewed, but by N=30 it already looks approximately normal, and at N=100 it closely resembles a perfect bell curve.

6.3. Does the sampling distribution become normal even though the original dataset are not normal?

Yes. This is exactly what the Central Limit Theorem states. The original Price data is heavily right-skewed, but the distribution of sample means becomes approximately normal regardless of the original shape, provided N is large enough.

6.4. What is the role of the standard error $\sigma/\sqrt{N}$?

It measures how much variability we expect in sample means from one sample to the next. A small SE means sample means are close to the true population mean μ , so estimates are reliable. As N grows, SE shrinks and our estimates improve.

6.5. Why does increasing N make the sample mean more reliable?

With a larger sample, extreme values have less influence on the mean. Averaging over more data points causes random highs and lows to cancel out, so the sample mean converges toward the true population mean μ .

6.6. How does the simulation confirm the CLT?

By repeating the sampling 1000 times for each N, we directly observe the predicted behaviour: the histogram of sample means becomes bell-shaped and narrower as N grows. The z-score histograms overlap the standard normal $N(0,1)$ curve more precisely as N increases, confirming the theoretical result computationally.

7-Conclusion:

In conclusion, this project helped us understand the Central Limit Theorem in a much more practical and visual way instead of just learning it theoretically. Even though the original housing-price data were uneven and heavily skewed, we observed that as the sample size increased, the distribution of the sample means gradually started looking more normal and consistent. We also noticed that larger sample sizes reduced the variation between sample means, making the estimates more reliable. By manually calculating the statistical values and repeatedly sampling the data, we were able to better understand how concepts like mean, standard deviation, standard error, and normalization work together in real-world data analysis. Overall, this experiment showed how powerful the Central Limit Theorem is and why it plays such an important role in statistics and data science.